

A One-Scale Legendre–Rvachev Closure on $[-1, 1]$

Exact Endpoint Synthesis, Appell Transmutation, and Thue–Morse Biorthogonality

Research report prepared from the PROVEIT Fabius–Rvachev corpus

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Abstract

Let $u = u_P$ be Rvachev’s even, compactly supported C^∞ function on $[-1, 1]$, normalized by $\int u = 1$, and let P_n denote the Legendre polynomial with $P_n(1) = 1$. The repository contains two ingredients that do not appear there in their fully composed form: an exact representation of an arbitrary degree- d polynomial on an interval by $d + 2$ shifts of one common dilate of u , and an absolutely and uniformly convergent Fourier–Legendre series

$$u(x) = \sum_{m \geq 0} \lambda_{2m} P_{2m}(x), \quad -1 \leq x \leq 1.$$

This report specializes the polynomial theorem directly to Legendre polynomials and then substitutes the resulting finite formulas into every Legendre partial sum.

For $d \geq n$, put $M = 2^d$ and $k_{d,r} = M - d - 1 + r$. We give an output-sensitive exact algorithm for rational coefficients $\beta_{d,n,r}$ such that

$$P_n(x) = \frac{1}{2^d} \sum_{r=0}^{d+1} \beta_{d,n,r} u\left(\frac{x+1-2k_{d,r}}{2^{d+1}}\right), \quad -1 \leq x \leq 1.$$

Taking $d = n$ gives the degree-matched $n + 2$ -atom common-scale form. More importantly, at the Fourier–Legendre cutoff N , taking one common value $d = 2N$ places all $P_0, P_2, \dots, P_{2N}$ in the same dictionary. If $S_N = \sum_{m=0}^N \lambda_{2m} P_{2m}$ and $B_{N,r} = \sum_{m=0}^N \lambda_{2m} \beta_{2N,2m,r}$, then

$$S_N(x) = 4^{-N} \sum_{r=0}^{2N+1} B_{N,r} u\left(\frac{x+1-2(4^N - 2N - 1 + r)}{2 \cdot 4^N}\right).$$

Consequently the right side converges uniformly to u on $[-1, 1]$. Each cutoff uses only $2N + 2$ atom occurrences at one scale, compared with $(N + 1)(N + 2)$ occurrences if the Legendre terms are synthesized separately at their degree-matched scales.

A second contribution is a direct connection theory between Legendre polynomials and the centered Rvachev–Appell polynomials A_n , defined by $e^{xz}/M_u(z) = \sum A_n(x) z^n/n!$. We derive mutually inverse triangular matrices

$$P_{2m} = \sum_{r=0}^m C_{m,r} A_{2r}, \quad A_{2r} = \sum_{m=0}^r D_{r,m} P_{2m},$$

with entries expressed through moments, reciprocal moments, Bernoulli cumulants, complete Bell polynomials, and dyadic Fabius values. Their finite connection determinant is the universal product

$$\det C^{(N)} = \prod_{m=0}^N 2^{-2m} \binom{4m}{2m},$$

independent of the lower moment data. Conjugating the classical Legendre recurrence by $M_u(D)$ produces a nonlocal Bernoulli-deformed Legendre operator with position variable $\mathcal{X} = x + (\log M_u)'(D)$.

Finally, adjoining the finite Thue–Morse polynomiality certificate to the matrix of Legendre moments of the endpoint atoms gives an invertible $(d+2) \times (d+2)$ matrix. Its inverse simultaneously contains every Legendre-to-up coefficient vector and a canonically normalized nonpolynomial defect. This identifies the endpoint synthesis as a biorthogonal completion of Legendre moments by one Thue–Morse row. Exact computations verify all matrix inversions, recurrences, rank statements, and certificates through degree 24 for the Appell connection and through degree 12 for the one-scale atom formulas. New conjectures concern strict sign alternation, quadratic logarithmic coefficient growth, total positivity, a q -hypergeometric compression, and Lambert– W edge asymptotics.

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1 Problem, corpus audit, and nonduplication boundary

1.1 The requested composition

The task asks for a composition of two exact maps:

Legendre polynomial $\mapsto$ finite common-scale up-atom vector $\mapsto$ closed representation of u .

The first arrow is a specialization of a theorem for arbitrary polynomials. The second arrow uses the existing even Fourier–Legendre expansion of u . No interpolation grid is required. This direct route is important because a companion Lagrange–cardinal analysis already factors polynomial synthesis through Appell–Vandermonde matrices and a multiscale two-atom ladder. The present report avoids repeating that construction. Its central normal form is instead the fixed-scale endpoint dictionary, and its new structural objects are the Legendre–Appell connection, the transmuted Legendre operator, and the augmented Legendre-moment/Thue–Morse matrix.

1.2 Repository-wide thematic audit

The live docs hierarchy and its draft manifests were searched across all LaTeX branches for the relevant structures. The full tree was used as a thematic audit; detailed formula-by-formula reading concentrated on the following paths, which contain the ingredients actually used in proofs:

Repository path	Role in this report
Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex	Normalization, sinc product, exact moments, dyadic Fabius values, Fourier–Legendre expansion, and endpoint asymptotics.
semi-formalized-research-frontiers/drafts/representations/Up_Polynomial_Synthesis/	Global polynomial reproduction, complete dyadic alias nullspace, the q -integer annihilator, the $d + 2$ endpoint basis, and the output-sensitive H_d algorithm.
semi-formalized-research-frontiers/drafts/inverse-and-sampling/	Rvachev–Appell polynomials, moment inversion, exact sampling and dyadic comb structures.
semi-formalized-research-frontiers/drafts/representations/	Representation questions, shift-invariant spaces, defect directions, and finite atomic extensions.
semi-formalized-research-frontiers/drafts/exponents-and-q-series/	q -Pochhammer and Gaussian-binomial formulas for dyadic special values.
semi-formalized-research-frontiers/drafts/integration-and-transforms/	Polynomial-weighted integration and exact reduction of dyadic cell integrals.
Non_Elementarity_of_the_Fabius_Function/	The nowhere-analytic/nonpolynomial obstruction used in the local dimension theorem.

The corpus also contains broad frontier compilations involving inverse Fabius asymptotics, Lambert– W , Bell and Bernoulli polynomials, Thue–Morse products, half-integer spectra, and generalized bases. Those materials are used below to frame conjectures, but they are not silently promoted to proved statements.

1.3 Status convention

Throughout the report:

- *imported* means proved in the audited repository corpus;
- *proved here* means a proof is supplied in this report;
- *verified exactly* means a finite symbolic check was performed by the accompanying program, in addition to the mathematical proof where one is claimed;
- *numerical* and *conjectural* are stated explicitly.

The novelty claims are relative to the audited repository snapshot and the companion reports available for this project; no unqualified claim of global priority is made.

2 Imported Rvachev, Fabius, and Legendre data

2.1 The up-function and its transforms

We use $u = \text{up}$ normalized by

$$u \in C_c^\infty(\mathbb{R}), \quad \text{supp } u = [-1, 1], \quad u(-x) = u(x), \quad \int_{-1}^1 u(x) \, dx = 1. \quad (1)$$

With Fourier convention $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} \, dx$, the transform is

$$\widehat{u}(\xi) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi \xi}{2^j}\right), \quad \text{sinc } z = \frac{\sin z}{z}. \quad (2)$$

At a nonzero integer m ,

$$\text{ord}_{\xi=m} \widehat{u}(\xi) = 1 + v_2(m). \quad (3)$$

This multiplicity is the spectral reason that the minimal dyadic dilation reproducing all polynomials through degree d is 2^d .

Let

$$M_u(z) = \int_{-1}^1 e^{zx} u(x) \, dx = \prod_{j=1}^{\infty} \frac{\sinh(z/2^j)}{z/2^j} = \sum_{n \geq 0} \mu_n \frac{z^n}{n!}. \quad (4)$$

Its logarithm is

$$\log M_u(z) = \sum_{m \geq 1} \kappa_{2m} \frac{z^{2m}}{(2m)!}, \quad \boxed{\kappa_{2m} = \frac{B_{2m}}{2m(1 - 4^{-m})}}, \quad (5)$$

where B_n are Bernoulli numbers. Hence

$$\mu_n = \mathcal{B}_n(\kappa_1, \dots, \kappa_n), \quad \gamma_n = \mathcal{B}_n(-\kappa_1, \dots, -\kappa_n), \quad (6)$$

for the complete exponential Bell polynomials, where

$$\frac{1}{M_u(z)} = \sum_{n \geq 0} \gamma_n \frac{z^n}{n!}. \quad (7)$$

The first values are

$$(\mu_0, \mu_2, \mu_4, \mu_6, \mu_8) = \left(1, \frac{1}{9}, \frac{19}{675}, \frac{583}{59535}, \frac{132809}{32531625}\right), \quad (8)$$

$$(\gamma_0, \gamma_2, \gamma_4, \gamma_6, \gamma_8) = \left(1, -\frac{1}{9}, \frac{31}{675}, -\frac{2347}{59535}, \frac{1904369}{32531625}\right). \quad (9)$$

All odd entries vanish.

2.2 Fabius special values

On the left half of the support one has

$$u(s-1) = F(s), \quad 0 \leq s \leq 1, \quad (10)$$

where F is the Fabius function. The exact moment/special-value transfer is

$$\mu_{2m} = 2(2m)! 2^{\binom{2m+1}{2}} F(2^{-2m-1}). \quad (11)$$

Thus every formula below that uses μ_{2m} can equally be written using dyadic Fabius values. The repository's q -binomial formulas for those special values then turn the same coefficients into finite q -hypergeometric data at $q = 1/2$.

2.3 The even Fourier–Legendre series

Let P_n be normalized by $P_n(1) = 1$. Since u is even,

$$u(x) = \sum_{m=0}^{\infty} \lambda_{2m} P_{2m}(x), \quad \lambda_{2m} = \frac{4m+1}{2} \int_{-1}^1 u(x) P_{2m}(x) dx. \quad (12)$$

The series is absolutely and uniformly convergent on $[-1, 1]$ in the primary repository exposition. The exact dyadic coefficient formula is

$$\begin{aligned} \lambda_{2m} = 4^{-m} (4m+1) \sum_{j=0}^m (-1)^{m+j} \binom{2m}{m+j} \binom{2m+2j}{2m} (2j)! \\ \times 2^{\binom{2j+1}{2}} F(2^{-2j-1}). \end{aligned} \quad (13)$$

In particular,

$$\lambda_0 = \frac{1}{2}, \quad \lambda_2 = -\frac{5}{6}, \quad \lambda_4 = \frac{11}{30}, \quad \lambda_6 = \frac{143}{5670}. \quad (14)$$

Write

$$S_N(x) = \sum_{m=0}^N \lambda_{2m} P_{2m}(x). \quad (15)$$

Then $S_N \rightarrow u$ uniformly.

3 The imported arbitrary-polynomial endpoint theorem

3.1 The common-scale dictionary

Fix $d \geq 0$, put

$$M = 2^d, \quad k_{d,r} = M - d - 1 + r, \quad 0 \leq r \leq d+1, \quad (16)$$

and define on $0 \leq t \leq 1$

$$\psi_{d,r}(t) = \frac{1}{M} u\left(\frac{t - k_{d,r}}{M}\right). \quad (17)$$

The repository proves that these $d+2$ functions form a basis of the local shift space generated by all active M -scale translates, that the polynomial part of this space is exactly Π_d , and that every $P \in \Pi_d$ has a unique representation

$$P(t) = \frac{1}{M} \sum_{r=0}^{d+1} b_r u\left(\frac{t - k_{d,r}}{M}\right), \quad 0 \leq t \leq 1. \quad (18)$$

If $P \in \mathbb{Q}[t]$, then $b_r \in \mathbb{Q}$. Moreover, $d + 2$ is the least possible size of any fixed dictionary of nontrivial affine up-atoms whose span is required to contain all of Π_d . This fixed-dictionary result does not assert that every individual target polynomial needs $d + 2$ atoms when shifts, scales, and coefficients may all depend nonlinearly on the target.

3.2 The q -integer annihilator

Define

$$\mathfrak{A}_d(z) = \prod_{m=1}^d [2^m]_z, \quad [q]_z = 1 + z + \cdots + z^{q-1}, \quad \mathfrak{A}_d(z) = \sum_{j \geq 0} a_{d,j} z^j. \quad (19)$$

Then

$$\mathfrak{A}_d(1) = 2^{d(d+1)/2}. \quad (20)$$

The full annihilator has exponentially large degree $2^{d+1} - d - 2$, but only $a_{d,0}, \dots, a_{d,d+1}$ are needed for the sparse endpoint coordinates.

3.3 The Bernoulli output-sensitive multiplier

Define

$$H_d(z) = \left(\frac{z}{2 \sinh(z/2)} \right)^d \frac{1}{M_u(z)}. \quad (21)$$

Its logarithm is

$$\log H_d(z) = - \sum_{m \geq 1} \frac{B_{2m}}{2m(2m)!} \left(d + \frac{1}{1 - 4^{-m}} \right) z^{2m}. \quad (22)$$

Only terms through degree d act nontrivially on Π_d .

Algorithm 3.1 (Output-sensitive endpoint coordinates). Given $P \in \Pi_d$:

1. compute $G(t) = H_d(D_t)P(t)$, truncating every formal series at degree d ;
2. for $0 \leq r \leq d + 1$, set

$$g_r = 2^{d(d+1)/2} G\left(r - \frac{d}{2}\right); \quad (23)$$

3. solve the unit-triangular convolution

$$b_0 = g_0, \quad b_r = g_r - \sum_{s=0}^{r-1} a_{d,r-s} b_s. \quad (24)$$

Then the b_r satisfy (18). The arithmetic cost is $\mathcal{O}(d^2)$ with quadratic truncated-series operations.

3.4 The finite Thue–Morse certificate

Let

$$\varepsilon_j = (-1)^{s_2(j)}, \quad (25)$$

where $s_2(j)$ is the binary digit sum. The factorization

$$\mathfrak{A}_d(z)(z - 1)^{d+1} = (-1)^{d+1} (1 - z) \sum_{j=0}^{2^d-1} \varepsilon_j z^{2j} \quad (26)$$

turns the complete alias recurrence into one signed first-difference filter. Restricted to the endpoint basis, a convenient certificate row is

$$\tau_{d,r} = (-1)^{r+1} \varepsilon_{\lfloor (2^{d+1}-d-2+r)/2 \rfloor}, \quad 0 \leq r \leq d+1. \quad (27)$$

The endpoint synthesis $\sum b_r \psi_{d,r}$ is polynomial if and only if

$$\sum_{r=0}^{d+1} \tau_{d,r} b_r = 0. \quad (28)$$

A nonzero scalar multiple of τ_d gives the same hyperplane.

4 Direct finite synthesis of Legendre polynomials

4.1 Affine specialization to $[-1, 1]$

For $d \geq n$, set

$$\tilde{P}_n(t) = P_n(2t - 1), \quad 0 \leq t \leq 1. \quad (29)$$

Apply Algorithm 3.1 to $\tilde{P}_n$ and denote the output by $\beta_{d;n,r}$.

Theorem 4.1 (Direct Legendre-to-up endpoint transform). *For integers $d \geq n \geq 0$, let*

$$\tilde{G}_{d,n}(t) = H_d(D_t) P_n(2t - 1). \quad (30)$$

Define

$$g_{d;n,r} = 2^{d(d+1)/2} \tilde{G}_{d,n} \left(r - \frac{d}{2} \right), \quad 0 \leq r \leq d+1, \quad (31)$$

and recursively

$$\beta_{d;n,0} = g_{d;n,0}, \quad \beta_{d;n,r} = g_{d;n,r} - \sum_{s=0}^{r-1} a_{d,r-s} \beta_{d;n,s}. \quad (32)$$

Then, for $-1 \leq x \leq 1$,

$$P_n(x) = \frac{1}{2^d} \sum_{r=0}^{d+1} \beta_{d;n,r} u \left(\frac{x+1-2(2^d-d-1+r)}{2^{d+1}} \right). \quad (33)$$

Every coefficient is rational. For fixed d , the vectors $\beta_{d;0}, \dots, \beta_{d;d}$ are linearly independent and span the polynomiality hyperplane $\tau_d^\perp \subset \mathbb{Q}^{d+2}$.

Proof. Apply (18) to (29). Under $t = (x+1)/2$,

$$\frac{t - k_{d,r}}{2^d} = \frac{x+1-2k_{d,r}}{2^{d+1}},$$

which gives (33). The coefficient algorithm is exactly Algorithm 3.1. Rationality follows from (22), Bernoulli rationality, and the integer triangular filter. The P_n form a basis of Π_d ; uniqueness of endpoint coordinates makes their vectors linearly independent. By (28) they lie in $\tau_d^\perp$, which has dimension $d+1$, so they span it. $\square$

Corollary 4.2 (Degree-matched formula). *Taking $d = n$ gives a finite formula with $n+2$ shifts of one common dilate of u . This dictionary is optimal among fixed dictionaries spanning all of Π_n .*

Remark 4.3 (What “shifted copies” means). For a fixed d , every summand in (33) is a translate of the same dilated function $x \mapsto u(x/2^{d+1})$. Thus the formula is a finite linear combination of shifts at one scale, rather than a collection of unrelated affine rescalings.

4.2 Low-degree formulas

For $P_0 = 1$, (33) becomes the partition identity

$$P_0(x) = u\left(\frac{x+1}{2}\right) + u\left(\frac{x-1}{2}\right). \quad (34)$$

For $P_2(x) = (3x^2 - 1)/2$, the degree-matched scale is $2^3 = 8$ and

$$P_2(x) = \frac{68}{3}u\left(\frac{x-1}{8}\right) - \frac{140}{3}u\left(\frac{x-3}{8}\right) + \frac{140}{3}u\left(\frac{x-5}{8}\right) - \frac{68}{3}u\left(\frac{x-7}{8}\right). \quad (35)$$

The next case already exposes the severe boundary cancellation. With the coefficient in the second column multiplying $u((x-c)/32)$, The coefficients are large because the arguments of u lie in a thin

Table 2: Degree-matched one-scale representation of P_4 .

c	amplitude
21	13179776/135
23	-17710976/45
25	93960704/135
27	-108475904/135
29	24331136/27
31	-147093632/135

layer near the flat endpoint -1 . The identity itself is exact; the difficulty is numerical evaluation, not algebraic validity.

4.3 A signed-coordinate phenomenon

Exact computation through P_{12} gives strict alternating signs in the right-endpoint coordinate vector:

m	degree	sign pattern of β_m	strict alternation
0	0	++	n/a
1	2	+--+	yes
2	4	+-+-+-	yes
3	6	+-+-+--	yes
4	8	+-+-+--+	yes
5	10	+-+-+--+--	yes
6	12	+-+-+--+--+	yes

This observation will become Conjecture 12.1. It is not a consequence of parity alone: the endpoint basis is geometrically asymmetric, and an even polynomial need not have a symmetric coordinate vector.

5 Closing the Fourier–Legendre loop at one scale

5.1 Common dictionary at cutoff N

At cutoff N , choose

$$d = 2N, \quad M_N = 2^{2N} = 4^N, \quad k_{N,r} = M_N - 2N - 1 + r, \quad 0 \leq r \leq 2N + 1. \quad (36)$$

Let

$$\beta_{m,r}^{(N)} = \beta_{2N;2m,r}, \quad 0 \leq m \leq N, \quad (37)$$

so that all even Legendre polynomials through degree $2N$ use exactly the same atoms.

Theorem 5.1 (Finite one-scale Legendre–Rvachev closure). *Define*

$$B_{N,r} = \sum_{m=0}^N \lambda_{2m} \beta_{m,r}^{(N)}. \quad (38)$$

Then the N th Fourier–Legendre partial sum satisfies the exact identity

$$S_N(x) = 4^{-N} \sum_{r=0}^{2N+1} B_{N,r} u\left(\frac{x+1-2(4^N-2N-1+r)}{2 \cdot 4^N}\right), \quad -1 \leq x \leq 1. \quad (39)$$

Moreover,

$$\sum_{r=0}^{2N+1} \tau_{2N,r} B_{N,r} = 0. \quad (40)$$

Proof. Use Theorem 4.1 with the common value $d = 2N$ for every P_{2m} in (15). Interchanging the two finite sums gives (39) and (38). Equation (40) follows either from linearity and the individual certificates, or because S_N is a polynomial of degree at most $2N$. $\square$

Corollary 5.2 (Infinite closed loop). *For $-1 \leq x \leq 1$,*

$$u(x) = \lim_{N \rightarrow \infty} 4^{-N} \sum_{r=0}^{2N+1} B_{N,r} u\left(\frac{x+1-2(4^N-2N-1+r)}{2 \cdot 4^N}\right) \quad (41)$$

with uniform convergence in x .

Proof. The finite sum is identically S_N by Theorem 5.1, and the imported Fourier–Legendre theorem gives $S_N \rightarrow u$ uniformly. $\square$

Warning 5.3 (No termwise limiting claim). The dictionary, scale, centers, and coefficients all depend on N . Uniform convergence in (41) is inherited from the polynomial side. It does not imply coordinatewise convergence of $B_{N,r}$, absolute convergence of a fixed atom series, or numerical stability of the displayed sum.

5.2 The first two loops

For $N = 0$,

$$S_0(x) = \frac{1}{2}u\left(\frac{x+1}{2}\right) + \frac{1}{2}u\left(\frac{x-1}{2}\right) = \frac{1}{2}. \quad (42)$$

For $N = 1$, using $S_1 = \frac{1}{2} - \frac{5}{6}P_2$,

$$\begin{aligned} S_1(x) = & -\frac{161}{9}u\left(\frac{x-1}{8}\right) + \frac{341}{9}u\left(\frac{x-3}{8}\right) \\ & -\frac{341}{9}u\left(\frac{x-5}{8}\right) + \frac{161}{9}u\left(\frac{x-7}{8}\right). \end{aligned} \quad (43)$$

Even at this first nonconstant cutoff, four large boundary terms cancel to produce a polynomial whose size is of order one.

5.3 Atom-count comparison

There are three natural finite normal forms at cutoff N .

1. Synthesizing each P_{2m} in its own degree-matched endpoint dictionary uses

$$\sum_{m=0}^N (2m+2) = (N+1)(N+2) \quad (44)$$

atom occurrences across $N+1$ scales.

2. The earlier two-atom Appell ladder uses $2(2N+1) = 4N+2$ occurrences across exponentially separated scales.
3. The present common-scale collapse uses $2N+2$ occurrences at one scale.

The third count is optimal for a fixed dictionary required to span all of Π_{2N} . It is not asserted to be target-adaptively minimal for the single even polynomial S_N , nor to be minimal among parity-adapted blocks.

One-scale compression principle. The nominal double sum obtained by atomizing every Legendre term separately contains $(N+1)(N+2)$ occurrences. Re-embedding all terms into the common maximal-degree dictionary and using linearity compresses the whole cutoff to $2N+2$ distinct atoms without approximation.

6 A direct Legendre–Rvachev–Appell connection

6.1 Centered Rvachev–Appell polynomials

Define A_n by

$$\frac{e^{xz}}{M_u(z)} = \sum_{n=0}^{\infty} A_n(x) \frac{z^n}{n!}. \quad (45)$$

Then

$$A'_n(x) = nA_{n-1}(x), \quad M_u(D)A_n(x) = x^n, \quad (46)$$

and, because M_u is even, A_n has the parity of n .

6.2 Forward connection coefficients

Write the even Legendre polynomial as

$$P_{2m}(x) = \sum_{j=0}^m p_{m,j} x^{2j}, \quad p_{m,j} = (-1)^{m-j} \frac{(2m+2j)!}{2^{2m}(m-j)!(m+j)!(2j)!}. \quad (47)$$

Theorem 6.1 (Legendre-to-Rvachev–Appell transform). *For every $m \geq 0$,*

$$P_{2m}(x) = \sum_{r=0}^m C_{m,r} A_{2r}(x), \quad (48)$$

where

$$C_{m,r} = \sum_{j=r}^m p_{m,j} \frac{(2j)!}{(2r)!} \frac{\mu_{2(j-r)}}{(2(j-r))!} \quad (49)$$

$$= \frac{1}{2^{2m}(2r)!} \sum_{j=r}^m (-1)^{m-j} \frac{(2m+2j)!}{(m-j)!(m+j)!(2(j-r))!} \mu_{2(j-r)}. \quad (50)$$

Equivalently, using (11), every $C_{m,r}$ is a finite linear combination of the dyadic values

$$F(2^{-1}), \quad F(2^{-3}), \quad \dots, \quad F(2^{-2(m-r)-1}).$$

Proof. Let

$$Q_{2m}(x) = M_u(D)P_{2m}(x). \quad (51)$$

If $Q_{2m}(x) = \sum_r C_{m,r}x^{2r}$, then applying $M_u(D)^{-1}$ and using (46) gives (48). Applying $M_u(D) = \sum_s \mu_s D^s / s!$ to the monomial expansion (47) yields (49); odd moments vanish. $\square$

The first rows are generated exactly in the accompanying code:

$$\begin{aligned} P_0(x) &= (1) A_0(x) \\ P_2(x) &= \left(-\frac{1}{3}\right) A_0(x) + \left(\frac{3}{2}\right) A_2(x) \\ P_4(x) &= \left(\frac{11}{135}\right) A_0(x) + \left(-\frac{5}{6}\right) A_2(x) + \left(\frac{35}{8}\right) A_4(x) \\ P_6(x) &= \left(\frac{11}{2835}\right) A_0(x) + \left(-\frac{7}{15}\right) A_2(x) + \left(\frac{35}{8}\right) A_4(x) + \left(\frac{231}{16}\right) A_6(x) \end{aligned}$$

Several low coefficients coincide with the Fourier–Legendre coefficients of u ; for example $C_{2,0} = 11/135$ while $\lambda_4 = 11/30$. This is not a general equality, but both arrays are different contractions of the same moment sequence.

6.3 Inverse connection coefficients

The Appell polynomials have the direct monomial form

$$A_{2r}(x) = \sum_{s=0}^r \binom{2r}{2s} \gamma_{2s} x^{2r-2s}. \quad (52)$$

The inverse monomial-to-Legendre coefficient is

$$x^{2q} = \sum_{m=0}^q L_{q,m} P_{2m}(x), \quad L_{q,m} = \frac{(4m+1)2^{2m}(2q)!(q+m)!}{(q-m)!(2q+2m+1)!}. \quad (53)$$

Theorem 6.2 (Inverse Rvachev–Appell-to-Legendre transform). *For $r \geq 0$,*

$$A_{2r}(x) = \sum_{m=0}^r D_{r,m} P_{2m}(x), \quad (54)$$

where

$$D_{r,m} = \sum_{s=0}^{r-m} \binom{2r}{2s} \gamma_{2s} \frac{(4m+1)2^{2m}(2r-2s)!(r-s+m)!}{(r-s-m)!(2r-2s+2m+1)!}. \quad (55)$$

For every N , the lower-triangular matrices

$$C^{(N)} = (C_{m,r})_{0 \leq r \leq m \leq N}, \quad D^{(N)} = (D_{r,m})_{0 \leq m \leq r \leq N} \quad (56)$$

satisfy

$$C^{(N)} D^{(N)} = D^{(N)} C^{(N)} = I_{N+1}. \quad (57)$$

Proof. Insert (53) into (52); the coefficient of P_{2m} is (55). Equations (48) and (54) are inverse changes of basis, hence (57). $\square$

6.4 A universal determinant

Theorem 6.3 (Universal connection determinant). *For every $N \geq 0$,*

$$\det C^{(N)} = \prod_{m=0}^N 2^{-2m} \binom{4m}{2m}. \quad (58)$$

In particular, the determinant is independent of all subleading Rvachev moments. More generally, the same determinant holds for the connection from even Legendre polynomials to the Appell family associated with any normalized even formal moment series $M(z)$ with $M(0) = 1$.

Proof. The differential operator $M_u(D)$ preserves the leading coefficient of a polynomial. Therefore the diagonal entry $C_{m,m}$ is the leading coefficient of P_{2m} ,

$$C_{m,m} = 2^{-2m} \binom{4m}{2m}. \quad (59)$$

The determinant of a triangular matrix is the product of its diagonal entries. The argument uses only $M(0) = 1$, proving universality. $\square$

Corollary 6.4 (Determinant asymptotics). *As $N \rightarrow \infty$,*

$$\log \det C^{(N)} = (\log 2)N(N+1) - \frac{N}{2} \log(2\pi) - \frac{1}{2} \log(N!) - \frac{1}{16} H_N + \mathcal{O}(1) \quad (60)$$

$$= (\log 2)N^2 - \frac{1}{2} N \log N + \mathcal{O}(N). \quad (61)$$

Consequently

$$\log \|C^{(N)}\|_2 \geq N \log 2 - \frac{1}{2} \log N + \mathcal{O}(1). \quad (62)$$

Proof. Use

$$2^{-2m} \binom{4m}{2m} = \frac{2^{2m}}{\sqrt{2\pi m}} \left(1 - \frac{1}{16m} + \mathcal{O}(m^{-2}) \right) \quad (63)$$

inside (58) and sum logarithms. The norm bound follows from $\|C\|_2^{N+1} \geq |\det C|$. $\square$

7 A Bernoulli-deformed Legendre operator

7.1 Conjugating multiplication by x

Set

$$K(z) = \log M_u(z), \quad \mathcal{X} = x + K'(D). \quad (64)$$

Since

$$M_u(D)x = xM_u(D) + M'_u(D), \quad (65)$$

we have the conjugation identity

$$M_u(D)xM_u(D)^{-1} = x + \frac{M'_u(D)}{M_u(D)} = \mathcal{X}. \quad (66)$$

Explicitly,

$$K'(D) = \sum_{j \geq 1} \kappa_{2j} \frac{D^{2j-1}}{(2j-1)!}, \quad (67)$$

so the deformation is governed term-by-term by Bernoulli numbers. The canonical commutator remains

$$[D, \mathcal{X}] = 1. \quad (68)$$

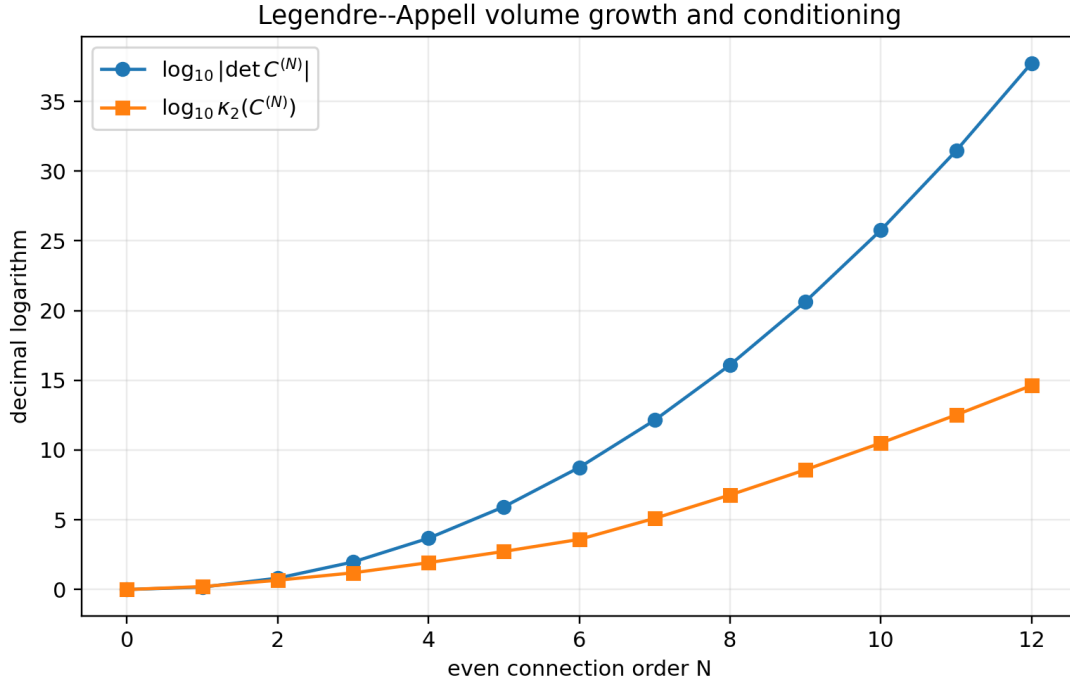


Figure 1: Exact determinant growth and floating-point spectral condition estimates for the even Legendre–Appell connection. The determinant is proved by [Theorem 6.3](#); condition numbers are diagnostic.

7.2 Transmuted recurrence and differential equation

Recall $Q_n = M_u(D)P_n$.

Theorem 7.1 (Rvachev-transmuted Legendre system). *The polynomials Q_n satisfy*

$$Q_0 = 1, \quad Q_1 = x, \quad (69)$$

and the operator recurrence

$$(n+1)Q_{n+1} = (2n+1)\mathcal{X}Q_n - nQ_{n-1}. \quad (70)$$

They also satisfy the nonlocal differential equation

$$[(1 - \mathcal{X}^2)D^2 - 2\mathcal{X}D + n(n+1)]Q_n = 0. \quad (71)$$

For even index,

$$Q_{2m}(x) = \sum_{r=0}^m C_{m,r} x^{2r}, \quad (72)$$

so (70) generates the connection matrix without separately expanding Legendre polynomials.

Proof. Apply $M_u(D)$ to the classical recurrence

$$(n+1)P_{n+1} = (2n+1)xP_n - nP_{n-1}$$

and use (66). Likewise conjugate the Legendre equation $[(1 - x^2)D^2 - 2xD + n(n+1)]P_n = 0$. Conjugation is an algebra homomorphism and D commutes with $M_u(D)$, giving (71). Finally, (72) is the definition of $C_{m,r}$ in the proof of [Theorem 6.1](#). $\square$

Remark 7.2 (Bispectral interpretation). The ordinary Legendre family is simultaneously controlled by a three-term recurrence in degree and a second-order differential operator in x . Theorem 7.1 preserves this architecture after replacing multiplication by x with the infinite-order operator $\mathcal{X}$. On each finite polynomial space the series truncates, so all identities are algebraic. A natural frontier question is whether this system admits a useful nonlocal orthogonality or a bispectral interpretation beyond formal conjugacy.

8 Legendre moments completed by a Thue–Morse row

8.1 Endpoint atoms on the physical interval

For fixed d , define

$$\Psi_{d,r}(x) = \frac{1}{2^d} u\left(\frac{x+1-2k_{d,r}}{2^{d+1}}\right), \quad 0 \leq r \leq d+1. \quad (73)$$

These functions form a basis of the $(d+2)$ -dimensional local up-space on $[-1, 1]$. Let

$$(\mathbf{L}_d)_{n,r} = \frac{2n+1}{2} \int_{-1}^1 \Psi_{d,r}(x) P_n(x) dx, \quad 0 \leq n \leq d. \quad (74)$$

Thus $\mathbf{L}_d \mathbf{b}$ records the first $d+1$ Legendre coefficients of the up-spline $\sum_r b_r \Psi_{d,r}$.

8.2 The augmented square transform

Define

$$\mathcal{Q}_d = \begin{pmatrix} \mathbf{L}_d \\ \tau_d^\top \end{pmatrix}, \quad (75)$$

where τ_d is (27).

Theorem 8.1 (Legendre–Thue–Morse biorthogonal completion). *The matrix $\mathcal{Q}_d \in \mathbb{R}^{(d+2) \times (d+2)}$ is invertible. If $\beta_{d;n}$ is the endpoint coordinate vector of P_n , then*

$$\mathcal{Q}_d \beta_{d;n} = \begin{pmatrix} \mathbf{e}_n \\ 0 \end{pmatrix}, \quad 0 \leq n \leq d. \quad (76)$$

Consequently the first $d+1$ columns of $\mathcal{Q}_d^{-1}$ are precisely $\beta_{d;0}, \dots, \beta_{d;d}$. The last column $\delta_d = \mathcal{Q}_d^{-1} \mathbf{e}_{d+1}$ is characterized by

$$\mathbf{L}_d \delta_d = 0, \quad \tau_d^\top \delta_d = 1. \quad (77)$$

Its synthesized function is the unique normalized nonpolynomial direction whose first $d+1$ Legendre moments vanish.

Proof. Suppose $\mathcal{Q}_d \mathbf{b} = 0$. The last row gives $\tau_d^\top \mathbf{b} = 0$, so the synthesis $f = \sum_r b_r \Psi_{d,r}$ is a polynomial in Π_d by the exact polynomiality certificate. The first $d+1$ rows say that every Legendre coefficient of f through degree d is zero. Hence $f = 0$. Because the $\Psi_{d,r}$ form a basis, $\mathbf{b} = 0$. Thus $\mathcal{Q}_d$ is invertible. For the endpoint vector of P_n , orthogonality and polynomiality give

$$\mathbf{L}_d \beta_{d;n} = \mathbf{e}_n, \quad \tau_d^\top \beta_{d;n} = 0,$$

which proves (76). The final assertion is the definition of the last inverse column. $\square$

Interpretation. The $d+1$ ordinary Legendre moment rows cannot distinguish the one-dimensional nonpolynomial defect inside the local $(d+2)$ -dimensional up-space. A single finite Thue–Morse row completes them to an invertible coordinate transform. The inverse contains both the complete polynomial synthesis table and the canonical defect coordinate.

8.3 Dyadic Fabius cell moments

For $d \geq 2$, all endpoint atoms in (73) sample the left half of the support. Set

$$a = d + 1 - r, \quad M = 2^d. \quad (78)$$

A change of variables followed by (10) gives

$$(\mathbb{L}_d)_{n,r} = (2n + 1) \int_{a/M}^{(a+1)/M} F(s) P_n(2Ms - (2a + 1)) \, ds. \quad (79)$$

Proof. In (74), put $y = (x + 1 - 2k_{d,r})/(2M)$. The factor M^{-1} in $\Psi_{d,r}$ and $dx = 2M \, dy$ leave a factor 2, which cancels the $1/2$ in the Legendre normalization. The integration interval is $[-k_{d,r}/M, (1 - k_{d,r})/M]$. With $s = y + 1$ it becomes $[a/M, (a + 1)/M]$, and $x = 2Ms - (2a + 1)$. $\square$

Equation (79) is a particularly concrete closure of the conceptual loop: the inverse of a matrix of adjacent dyadic Fabius cell moments, completed by one Thue–Morse row, gives the coefficients of Legendre polynomials in a finite up-atom dictionary. Polynomial expansion of P_n , followed by the repository’s polynomial-weighted integration identities, reduces these entries to exact dyadic Fabius data.

9 Arithmetic and q -structural consequences

9.1 Rationality at every layer

The following quantities are rational:

- the moments μ_{2m} and reciprocal moments γ_{2m} ;
- the connection coefficients $C_{m,r}$ and $D_{r,m}$;
- every endpoint coefficient $\beta_{d;n,r}$;
- the Fourier–Legendre coefficients λ_{2m} ;
- every finite closed-loop coefficient $B_{N,r}$.

The reasons are mutually reinforcing: Bernoulli cumulants generate moments and reciprocal moments by Bell recurrences; dyadic Fabius values are rational; the endpoint multiplier H_d has rational coefficients; and the q -integer triangular solve has integer entries and unit diagonal.

9.2 Three appearances of the same dyadic arithmetic

Three polynomial products occur in apparently different roles:

1. the infinite sinc product controls zero multiplicities through $1 + v_2(m)$;
2. the finite alias annihilator is the q -integer product $\mathfrak{A}_d(z) = \prod_{m=1}^d [2^m]_z$;
3. multiplying by $(z - 1)^{d+1}$ turns it into the finite Thue–Morse product (26).

The first chooses the scale 2^d , the second removes exponentially many alias degrees of freedom, and the third identifies the single obstruction to polynomiality after restriction. The same chain is present inside the Legendre formulas because Algorithm 3.1 is applied to $P_n(2t - 1)$ without alteration.

9.3 A q -binomial form of the connection coefficients

Combining (50) with (11) yields

$$C_{m,r} = \frac{2^{1-2m}}{(2r)!} \sum_{j=r}^m (-1)^{m-j} \frac{(2m+2j)!}{(m-j)!(m+j)!} \times 2^{\binom{2(j-r)+1}{2}} F\left(2^{-2(j-r)-1}\right). \quad (80)$$

The repository expresses the dyadic values in terms of Gaussian binomial and q -Pochhammer data. Substituting those formulas into (80) produces a finite $q = 1/2$ representation of the complete Legendre–Appell matrix. A useful unsolved problem is to resum the nested finite sums into a single terminating basic hypergeometric expression.

9.4 Denominators and valuations

The exact data show two distinct sources of denominator growth: Bernoulli/Bell denominators in H_d and the rational Legendre/Fabius moment coefficients. The q -integer solve itself introduces no division. This suggests studying odd-prime support separately from 2-adic valuation. Von Staudt–Clausen controls the Bernoulli primes, while the dyadic factors are governed by the explicit powers in (11) and by cancellation against $\mathfrak{A}_d(1) = 2^{d(d+1)/2}$.

10 Conditioning, endpoint flatness, and Lambert– W

10.1 Exact identities versus unstable evaluation

The endpoint formulas place all sample arguments in a boundary layer whose width is comparable with

$$\delta_d \asymp \frac{d}{2^d}. \quad (81)$$

Because $u(-1+s) = F(s)$ is flat at $s = 0$, the atom values are extremely small, whereas their rational coefficients are extremely large. Cancellation is therefore intrinsic to the right-endpoint basis.

The exact data through degree 12 are summarized in Figure 2. For the raw coefficient vector of the degree-matched P_d formula,

$$\log_{10} \max_r |\beta_{d;d,r}| = 2.27, 7.24, 13.86, 21.92, 31.43, 42.32 \quad (82)$$

for $d = 2, 4, 6, 8, 10, 12$, respectively.

10.2 Closed-loop coefficient growth

For the one-scale partial-sum coefficients $B_{N,r}$, exact computation gives

The atom count is linear, but coefficient complexity is not. At $N = 6$, only 14 atoms remain after compression, while the largest raw coefficient has about 41 decimal digits and the largest numerator or denominator requires 217 bits.

10.3 Why Lambert– W appears

The repository’s sharp endpoint analysis reduces the small- s behavior of $F(s)$ to a saddle equation whose inversion uses a branch of Lambert– W . At the endpoint scale (81),

$$\log \frac{1}{\delta_d} = d \log 2 - \log d + \mathcal{O}(1). \quad (83)$$

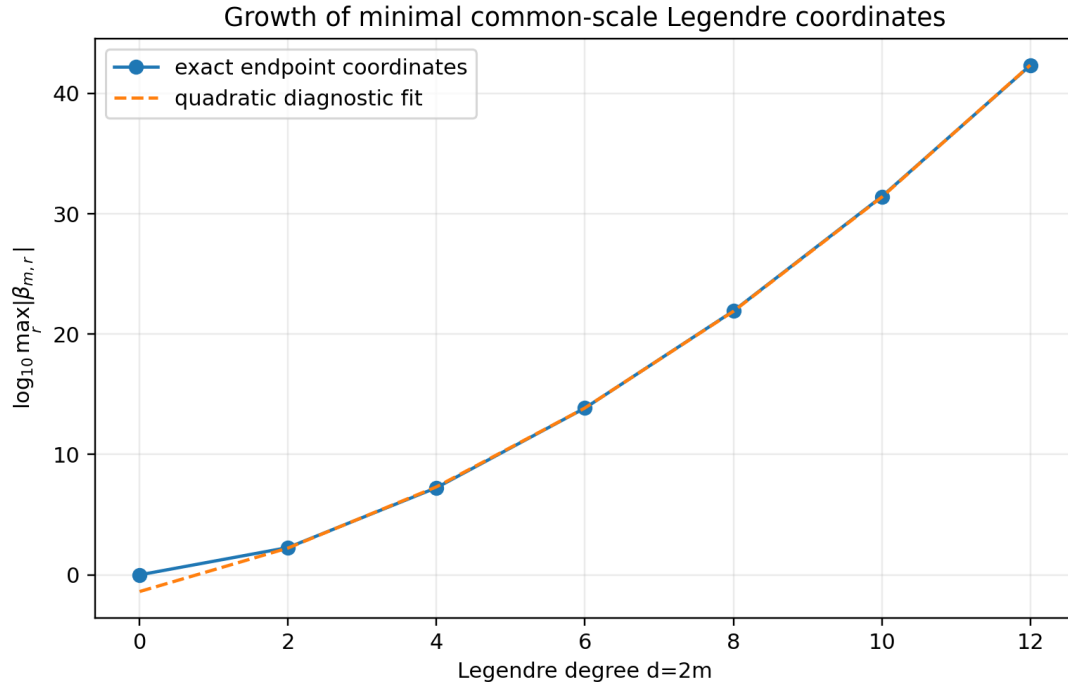


Figure 2: Exact rational endpoint-coordinate growth for the degree-matched even Legendre polynomials. The dashed curve is only a diagnostic quadratic fit, not a proved asymptotic.

Table 3: One-scale closed-loop coordinate growth. “Separate-scale occurrences” is $(N + 1)(N + 2)$.

N	degree	atoms	separate-scale occurrences	$\log_{10} \max B_{N,r} $	bits
0	0	2	2	-0.301	2
1	2	4	6	2.181	11
2	4	6	12	6.787	34
3	6	8	20	12.319	64
4	8	10	30	20.851	104
5	10	12	42	30.023	156
6	12	14	56	40.741	217

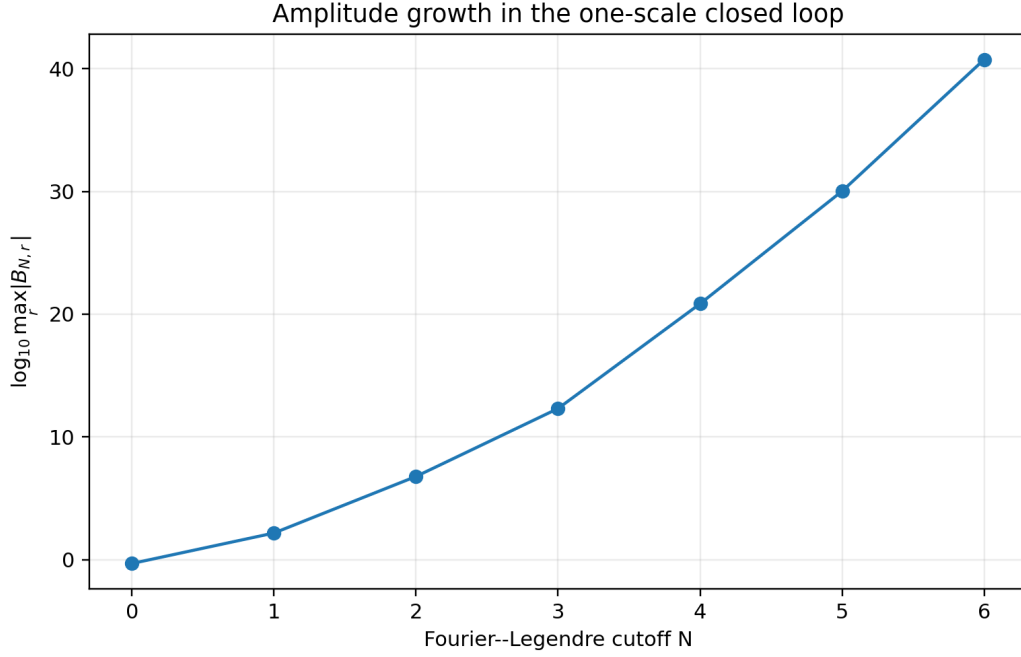


Figure 3: Maximum exact coordinate magnitude in the one-scale closed loop. The plotted values are decimal logarithms of exact rationals.

The leading flat-tail law is quadratic in this logarithm, with lower-order $\log \log(1/s)$ and periodic dyadic corrections. Independently, the triangular algorithm contains the mass $\mathfrak{A}_d(1) = 2^{d(d+1)/2}$. Heuristically, one quadratic contribution comes from this finite convolution law and another from compensating the flat endpoint values. This explains why the observed natural logarithm of the largest coefficient is close to a constant times d^2 , and motivates the sharp conjecture in [Theorem 12.2](#).

This heuristic must not be mistaken for a proof. A proof would require a uniform saddle analysis of a signed finite combination, not merely a pointwise asymptotic for F .

10.4 Numerical diagnostic

A finite-convolution approximation of u was used only to visualize the low-degree formulas. The maximum residuals on a 501-point grid were roughly 7.6×10^{-6} , 9.9×10^{-5} , and 1.2×10^{-3} for P_0, P_2, P_4 . The corresponding maximum cancellation ratios were about 1, 3.6×10^4 , and 4.7×10^8 .

11 Further deductions from the closed loop

11.1 A finite-rank extension/restriction factorization

Let R_d restrict a function to its first $d + 1$ Legendre coefficients, and let E_d map a polynomial coefficient vector to the endpoint up-spline with coordinates supplied by $\mathcal{Q}_d^{-1}$. Then on Π_d ,

$$R_d E_d = I. \tag{84}$$

Thus $E_d R_d$ is a finite-rank projection on the local up-space. The polynomial subspace is selected not by point sampling but by Legendre moments, and the extra defect coordinate is selected by Thue–Morse. This gives a basis-free interpretation of the endpoint construction.

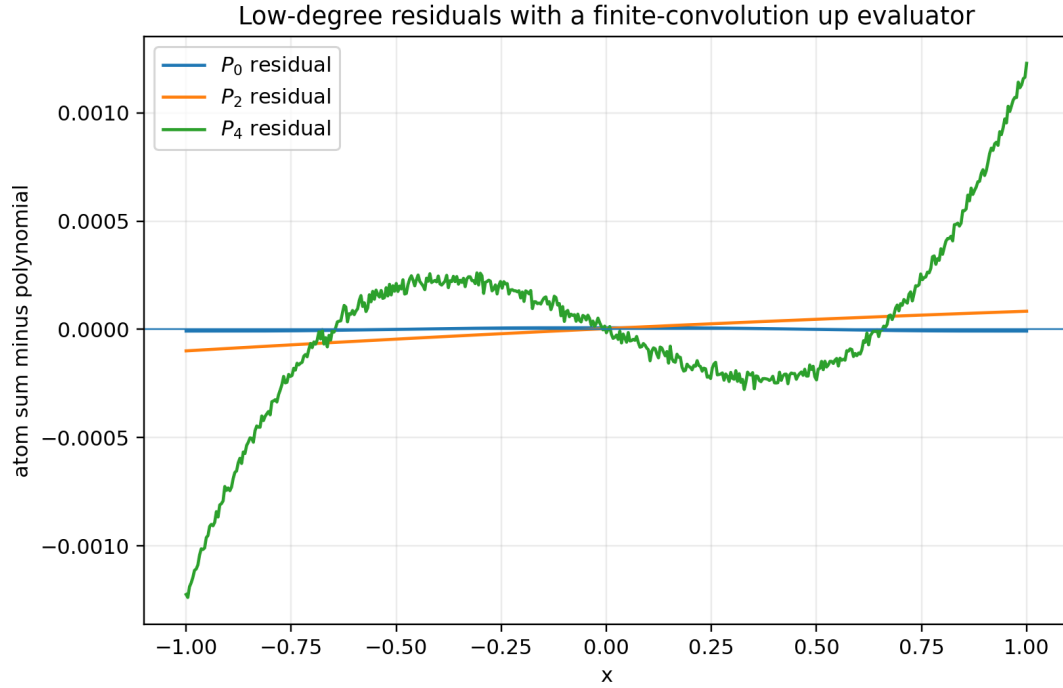


Figure 4: Floating-point residuals obtained from a finite-convolution approximation of u . The exact symbolic identities have zero residual; the plot measures evaluator and cancellation error.

11.2 A defect-sensitive reconstruction formula

For any local endpoint up-spline $f = \sum_r b_r \Psi_{d,r}$, define

$$\alpha_n = \frac{2n+1}{2} \int_{-1}^1 f(x) P_n(x) dx, \quad \eta = \tau_d^T \mathbf{b}. \quad (85)$$

Then

$$\mathbf{b} = \mathcal{Q}_d^{-1} \begin{pmatrix} \alpha_0 \\ \vdots \\ \alpha_d \\ \eta \end{pmatrix}. \quad (86)$$

Hence the local up-spline is determined by $d+1$ classical moments plus one digital certificate. Polynomial synthesis is the special case $\eta = 0$. This is an exact finite analogue of augmenting an incomplete moment problem by one boundary datum.

11.3 Universality beyond the Rvachev law

The direct connection theorem has a universal component. Given any normalized even moment series $M(z)$, define its Appell family by $e^{xz}/M(z)$. The formulas (49), (55), and (58) remain valid with the new moments and reciprocal moments. What is special to Rvachev is not the triangular Appell algebra but the finite realization of those Appell coordinates by compactly supported dyadic atoms, the q -integer alias annihilator, and the Thue–Morse defect row.

11.4 A route to fast coefficient generation

There are now two independent recurrences:

- (70) generates the Legendre–Appell rows $C_{m,r}$;

- (32) sends any polynomial row to the endpoint atom coordinates.

Their composition gives a structured transform from Legendre coefficients to endpoint coordinates without expanding each P_n in monomials. For a dense polynomial of degree d , both stages admit quadratic arithmetic complexity. The special three-term degree recurrence suggests that subquadratic algorithms may be possible using divide-and-conquer polynomial arithmetic and Toeplitz convolution.

12 Conjectures and frontier directions

Conjecture 12.1 (Strict endpoint alternation). *For every $m \geq 1$, the degree-matched right-endpoint coordinates of P_{2m} satisfy*

$$\boxed{(-1)^r \beta_{2m;2m,r} > 0, \quad 0 \leq r \leq 2m + 1.} \quad (87)$$

Exact computation verifies this through $m = 6$. A proof may follow from a total-positivity or extended-Chebyshev property of the endpoint atom system, but no such theorem is presently available for these flat nonanalytic atoms.

Conjecture 12.2 (Quadratic logarithmic endpoint growth). *Let $\mathfrak{B}_d = \max_r |\beta_{d;d,r}|$ for even d . Then*

$$\boxed{\lim_{\substack{d \rightarrow \infty \\ 2|d}} \frac{\log \mathfrak{B}_d}{d^2} = \log 2.} \quad (88)$$

The same leading constant is expected for $\max_r |B_{N,r}|$ after writing $d = 2N$, at least along a limsup. The proposed constant reflects the combined q -integer mass and flat-tail compensation; Lambert– W and dyadic periodic corrections should enter the lower-order terms.

Conjecture 12.3 (Legendre–Fabius augmented determinant). *After a natural diagonal normalization of the Legendre rows and the endpoint atoms, the determinant of $\mathcal{Q}_d$ factors into a product of powers of two, short q -integer factors, and a residual determinant with controlled odd prime support. Equivalently, the inverse table of all Legendre endpoint coordinates should admit a fraction-free elimination governed by the same cyclotomic factors as $\mathfrak{A}_d$.*

Conjecture 12.4 (Total positivity after checkerboard scaling). *There exists a canonical ordering and positive diagonal scaling of the right-endpoint atom evaluation or moment matrix for which checkerboard sign changes convert the relevant Legendre synthesis minors to nonnegative minors. Strict positivity of the maximal minors would imply Conjecture 12.1 and a variation-diminishing property for polynomial endpoint coordinates.*

Problem 12.5 (q -hypergeometric compression). *Insert the repository’s Gaussian-binomial formulas for $F(2^{-2s-1})$ into (80). Can the resulting double finite sum be expressed as one terminating ${}_r\phi_s$ series, a q -Racah-type connection coefficient, or a determinant of Gaussian binomial entries? Such a formula could expose hidden orthogonality in the index pair (m, r) .*

Problem 12.6 (Nonlocal orthogonality of the transmuted system). *Find a bilinear form, positive if possible, for which the polynomials $Q_n = M_u(D)P_n$ are orthogonal and the operator in (71) is symmetric. Formal conjugacy transports the Legendre inner product to*

$$\langle f, g \rangle_u = \int_{-1}^1 [M_u(D)^{-1}f](x) [M_u(D)^{-1}g](x) dx, \quad (89)$$

but locality, positivity on suitable completions, and a spectral theorem are nontrivial.

Problem 12.7 (Parity-adapted fixed dictionaries). *The even polynomial space of degree at most $2N$ has dimension $N + 1$, whereas the full endpoint dictionary has $2N + 2$ atoms. Can one construct $N + 1$ symmetric atom pairs or other parity-adapted blocks that form a fixed basis of the even subspace and improve conditioning? The count should be compared in blocks, raw atom occurrences, support overhang, and coefficient norm.*

Problem 12.8 (Condition-optimal common-scale basis). *Among all $(d + 2)$ -subsets of the active dyadic dictionary, minimize a stability functional such as*

$$\sup_{\|p\|_\infty \leq 1} \|\mathbf{b}(p)\|_1, \quad \kappa_2(\mathcal{Q}_d), \quad \text{or} \quad \max_r \text{dist}(\text{supp } \Psi_{d,r}, [-1, 1]). \quad (90)$$

The right-endpoint basis is algebraically canonical but strongly ill-conditioned. Central consecutive bases and symmetric hybrids are the first candidates.

Problem 12.9 (Sharp Lambert– W coefficient asymptotics). *Combine the exact triangular formula (32) with the repository’s sharp endpoint expansion for F , including its periodic dyadic correction. Determine*

$$\log \mathfrak{B}_d - (\log 2)d^2 \quad (91)$$

through the $d \log d$, d , $\log d$, and bounded periodic terms. The main technical obstacle is controlling cancellation among neighboring endpoint samples uniformly in the triangular solve.

Problem 12.10 (Two-adic and odd-prime valuations). *For fixed index ratios r/d , determine asymptotic or exact formulas for $v_2(\beta_{d;d,r})$ and for the odd denominator support. Separate the Bernoulli contribution, the Fabius moment contribution, and cancellation in the integer q -filter. The result should be compared with the prime support of Gaussian binomial coefficients at $q = 1/2$.*

Problem 12.11 (Beyond base two). *For base- b atomic functions, replace the sinc product, 2-adic zero multiplicity, and Thue–Morse product by their base- b analogues. Determine the corresponding Legendre endpoint dictionary, digit-character certificate, and Appell-transmuted recurrence. The determinant (58) remains universal, while the finite atomic realization should encode the new base.*

Problem 12.12 (Formal verification). *A compact formalization path is:*

1. *import the arbitrary-polynomial endpoint theorem and its H_d algorithm;*
2. *formalize the affine specialization in Theorem 4.1;*
3. *formalize the finite substitution in Theorem 5.1;*
4. *prove the moment formulas for C and D by finite sums;*
5. *verify the triangular determinant and the conjugated recurrence; and*
6. *formalize the kernel argument for Theorem 8.1.*

All steps after the imported analytic lemmas are finite polynomial or matrix algebra.

13 Conclusion

The arbitrary-polynomial endpoint theorem has a particularly clean Legendre specialization. Every P_n is exactly a finite linear combination of shifts of one common dyadic dilate of Rvachev’s up-function. At a Fourier–Legendre cutoff, all even Legendre terms can be moved to one maximal-degree dictionary, so the full partial sum collapses to $2N + 2$ atoms. Since the partial sums converge uniformly, this gives a closed self-representation of u by moving finite one-scale atom systems.

The direct Legendre–Appell transform reveals a second layer. Moment convolution sends P_{2m} to a polynomial whose monomial coefficients are the Appell coordinates. The forward and inverse matrices

are explicit finite Bernoulli–Bell transforms; their determinant is universal and depends only on the leading Legendre coefficients. Conjugating the Legendre operators by the moment series replaces x with the Bernoulli differential position operator $x + (\log M_u)'(D)$.

The endpoint coordinates admit a complementary moment interpretation. The matrix of the first $d + 1$ Legendre moments of the $d + 2$ endpoint atoms is one row short of being square. The missing row is exactly the finite Thue–Morse polynomiality certificate. Its inverse simultaneously produces all Legendre synthesis vectors and the unique normalized nonpolynomial defect. This is the most compact structural form of the loop: classical orthogonal moments are completed by one digital row arising from the same dyadic product that controls the sinc zeros and q -integer aliases.

The representation is exact but numerically delicate. Linear atom counts coexist with quadratic-logarithmic coefficient growth because the atoms are sampled near a flat endpoint. That is not a defect of the proof; it is a geometric feature connecting endpoint asymptotics, Lambert– W , Thue–Morse cancellation, and the arithmetic of dyadic Fabius values. The most promising next steps are a sharp saddle analysis of the coefficient envelope, a checkerboard total-positivity theorem, a condition-optimal central dictionary, and a q -hypergeometric formula for the Legendre–Appell matrix.

A Formula sheet

Moment and Appell data.

$$\begin{aligned} M_u(z) &= \prod_{j \geq 1} \frac{\sinh(z/2^j)}{z/2^j}, & \kappa_{2m} &= \frac{B_{2m}}{2m(1 - 4^{-m})}, \\ \frac{1}{M_u(z)} &= \sum_{n \geq 0} \gamma_n \frac{z^n}{n!}, & \frac{e^{xz}}{M_u(z)} &= \sum_{n \geq 0} A_n(x) \frac{z^n}{n!}. \end{aligned}$$

Endpoint dictionary.

$$\begin{aligned} M &= 2^d, & k_{d,r} &= M - d - 1 + r, \\ \Psi_{d,r}(x) &= 2^{-d} u\left(\frac{x + 1 - 2k_{d,r}}{2^{d+1}}\right), & 0 \leq r &\leq d + 1. \end{aligned}$$

Coefficient algorithm.

$$\begin{aligned} H_d(z) &= \left(\frac{z}{2 \sinh(z/2)}\right)^d \frac{1}{M_u(z)}, \\ \mathfrak{A}_d(z) &= \prod_{m=1}^d [2^m]_z = \sum_j a_{d,j} z^j, \\ g_r &= 2^{d(d+1)/2} [H_d(D)P](r - d/2), \\ b_0 &= g_0, & b_r &= g_r - \sum_{s < r} a_{d,r-s} b_s. \end{aligned}$$

Legendre endpoint formula.

$$P_n(x) = 2^{-d} \sum_{r=0}^{d+1} \beta_{d;n,r} u\left(\frac{x + 1 - 2(2^d - d - 1 + r)}{2^{d+1}}\right).$$

One-scale closed loop.

$$\begin{aligned}
B_{N,r} &= \sum_{m=0}^N \lambda_{2m} \beta_{2N;2m,r}, \\
S_N(x) &= 4^{-N} \sum_{r=0}^{2N+1} B_{N,r} u \left(\frac{x+1-2(4^N-2N-1+r)}{2 \cdot 4^N} \right), \\
u(x) &= \lim_{N \rightarrow \infty} S_N(x) \quad \text{uniformly on } [-1, 1].
\end{aligned}$$

Legendre–Appell connection.

$$\begin{aligned}
P_{2m} &= \sum_{r=0}^m C_{m,r} A_{2r}, & C_{m,r} &= [x^{2r}] M_u(D) P_{2m}, \\
A_{2r} &= \sum_{m=0}^r D_{r,m} P_{2m}, & \det C^{(N)} &= \prod_{m=0}^N 2^{-2m} \binom{4m}{2m}.
\end{aligned}$$

Transmuted Legendre system.

$$\begin{aligned}
Q_n &= M_u(D) P_n, & \mathcal{X} &= x + (\log M_u)'(D), \\
(n+1)Q_{n+1} &= (2n+1)\mathcal{X}Q_n - nQ_{n-1}, \\
[(1-\mathcal{X}^2)D^2 - 2\mathcal{X}D + n(n+1)]Q_n &= 0.
\end{aligned}$$

Augmented moment matrix.

$$\begin{aligned}
(\mathbb{L}_d)_{n,r} &= \frac{2n+1}{2} \int_{-1}^1 \Psi_{d,r} P_n, \\
\mathcal{Q}_d &= \begin{pmatrix} \mathbb{L}_d \\ \tau_d^\top \end{pmatrix}, & \beta_{d;n} &= \mathcal{Q}_d^{-1} \begin{pmatrix} e_n \\ 0 \end{pmatrix}.
\end{aligned}$$

B Pseudocode for the direct closed loop

Listing 1: Exact common-scale Legendre closure.

```

INPUT: cutoff N
OUTPUT: exact rational B[0..2N+1], centers, common scale

d <- 2N
M <- 2^d
construct moments mu[0..d] from Bernoulli cumulants
construct lambda[0..N] from integral up(x) P_(2m)(x)
construct H_d(z) modulo z^(d+1)
construct a[0..d+1] from product_(j=1)^d [2^j]_z,
    truncating every convolution

for m=0..N:
  P(t) <- P_(2m)(2t-1)
  G(t) <- H_d(D_t) P(t)
  for r=0..d+1:
    g[r] <- 2^(d(d+1)/2) G(r-d/2)
    beta[m,r] <- g[r] - sum_(s<r) a[r-s] beta[m,s]

```



```

for r=0..d+1:
  B[r] <- sum_(m=0)^N lambda[2m] beta[m,r]
  k[r] <- M-d-1+r
  center[r] <- 2k[r]-1

RETURN amplitude[r]=B[r]/M,
      atom[r](x)=up((x-center[r])/(2M))

equality: S_N(x)=sum_r amplitude[r] atom[r](x), -1<=x<=1

```

C Reproducibility files

The accompanying ZIP archive contains:

`legendre_rvachev_closed_loop.tex`

The complete LaTeX source.

`legendre_rvachev_closed_loop.pdf`

The rendered report.

`legendre_rvachev_experiments.py`

Commented exact and numerical experiments.

`data/legendre_appell_connection.csv`

Exact C and D entries through connection order 12.

`data/legendre_appell_conditioning.csv`

Exact determinants and floating-point condition estimates.

`data/minimal_legendre_endpoint_coefficients.csv`

Degree-matched endpoint vectors through P_{12} .

`data/fixed_scale_loop_summary.csv` **and** `data/fixed_scale_loop_coefficients.csv`

One-scale partial-sum coordinates through cutoff $N = 6$.

`data/thue_morse_certificate_checks.csv`

Exact certificate rows and zero residuals.

`data/transmuted_recurrence_checks.txt`

Exact zero residuals for the transmuted recurrence through $n = 13$.

`figures/` **and** `generated/`

The figures and exact LaTeX fragments used in the report.

`README.md`, `requirements.txt`, **and** `SHA256SUMS`

Reproduction instructions, dependencies, and integrity hashes.

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